

MATTER AND ATOMIC STRUCTURE

CHAPTER 1A

1. Zinc has a mass number of 66.
 - a. Determine the number of protons.
 - b. Determine the number of neutrons.
 - c. Determine the number of electrons.
2. Identify each description as a proton, neutrons, or electrons.
 - a. Has the smallest mass.
 - b. Carries +1 charge.
 - c. Is found outside the nucleus.
 - d. Is electrically neutral.
3. Write the symbols for the following species using the periodic table.
 - a. 19 protons, 18 electrons, 20 neutrons
 - b. 10 electrons, 8 neutrons, 8 protons
 - c. 126 neutrons, 82 protons, 80 electrons
 - d. 1 electron, 1 proton, 2 neutrons
4. An atom of element **A** contains the same neutron number as are found in an atom of ^{27}Al . An atom **A** also contains 14 protons.
 - a. Give the number of protons in an atom of ^{27}Al .
 - b. Deduce the symbol, including mass number and atomic number for element **A**.
5. Boron has two isotopes, one with five neutrons and the other with six neutrons.
 - a. Define the term *isotopes*.
 - b. Explain why isotopes of the same element have the same chemical properties.
 - c. Determine the mass number of these two isotopes.

6. Name the following ions:

No.	Formula of ions	Name of ions
1.	NH_4^+	
2.	NO_3^-	
3.	NO_2^-	
4.	ClO_4^-	
5.	ClO_3^-	
6.	ClO_2^-	
7.	ClO^-	
8.	CO_3^{2-}	
9.	HCO_3^{2-}	
10.	CN^-	
11.	SO_4^{2-}	
12.	HSO_4^-	
13.	SO_3^{2-}	
14.	HSO_3^-	
15.	PO_4^{3-}	
16.	HPO_4^{2-}	
17.	H_2PO_4^-	
18.	PO_3^{3-}	
19.	CrO_4^{2-}	
20.	$\text{Cr}_2\text{O}_7^{2-}$	
21.	MnO_4^-	

7. Name the following compounds:

No.	Formula	Name
1.	BaSO_4	
2.	CaCO_3	
3.	CaSO_3	
4.	CaSO_4	
5.	AgNO_3	
6.	NaHCO_3	
7.	$\text{Zn}_3(\text{PO}_4)_2$	
8.	FePO_4	
9.	K_2CO_3	
10.	$\text{Al}_2(\text{SO}_4)_3$	
11.	AlPO_4	
12.	MgSO_4	
13.	KClO_3	
14.	$\text{Mn}(\text{OH})_2$	
15.	Na_3P	
16.	Na_2SO_4	
17.	$\text{Cr}(\text{ClO})_3$	
18.	$\text{Cu}(\text{NO}_2)_2$	

8. Sulphur consists of three isotopes. The table below shows the relative abundance of each isotopes. Using the data, calculate the relative atomic mass of sulphur, giving your answer to one decimal place.

Mass number of isotopes	32	33	34
Relative abundance, %	95.0	0.8	4.2

9. Deduce the relative atomic masses of the following elements.

- a. Silicon (^{28}Si 92.21%, ^{29}Si 4.70%, ^{30}Si 3.09%)
- b. Gallium (^{69}Ga 60.2%, ^{71}Ga 39.8%)
- c. Boron (^{10}B 19.3%, ^{11}B 80.7%)
- d. Silver (^{107}Ag 51.88%, ^{109}Ag 48.12%)

10. A gaseous sample of chromium can be analyzed in a mass spectrometer. Before deflection, the chromium atoms are ionized and then accelerated.

- a. Describe briefly how positive ions are formed from gaseous chromium atoms in a mass spectrometer.
- b. What is used in a mass spectrometer to accelerate the positive ions?
- c. What is used in a mass spectrometer to deflect the positive ions?
- d. The mass spectrum of a chromium sample shows four peaks. Use the data below to calculate the relative atomic mass of chromium in the sample. Give your answer in two decimal places.

m/z	50	52	53	54
Relative Abundance, %	4.3	83.8	9.5	2.4

11. Isotopes can be separated in a mass spectrometer. Show how this is possible by describing the various parts of a mass spectrometer and by discussing the principles of operation of each part.

Answers:

1a) 30 b) 36 c) 30

4a) 14

8) 32.1 amu

9a) 28.11 amu b) 69.8 amu c) 10.8 amu d) 107.96 amu

10d) 52.06 amu